

Electrical Machines

Detailed Solutions • 2013–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Electrical Machines (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: IN	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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SOLUTIONS

Magnetic Circuits

Topics

1. Magnetically Coupled Circuits, Dot Convention
2. Energy in Coupled circuits
3. Electric and Magnetic Materials

Unit Snapshot*

Stream: IN	**Questions:** 6	**MCQ:** 4	**MSQ:** 0	**NAT:** 2
1M: 2	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2025)–4M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q1.25.1 MCQ 2025 1M Ans: A ●●○

Given circuit is in a magnetically opposing dot configuration. By modeling the magnetically coupled coils into its equivalent T-network:

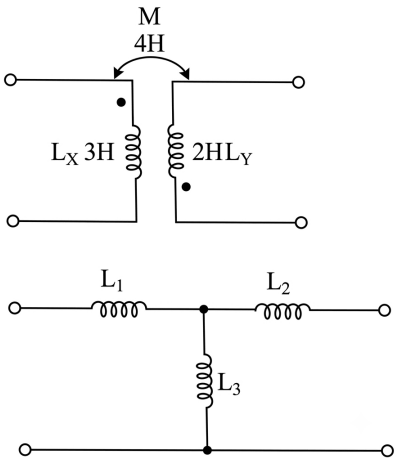
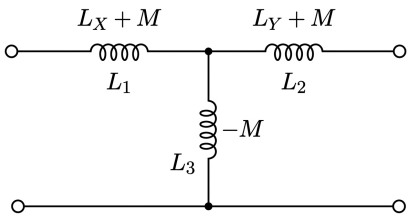


Fig. Solution Q1.25.1



Equivalent circuit

Fig. Solution Q1.25.1

For a mutually coupled inductor pairs with opposing dots, the mutual terms add up positively in the series arms of the equivalent T-network ($L_1 = L_X + M$, $L_2 = L_Y + M$) and negatively in the common shunt branch ($L_3 = -M$). The values of the equivalent T-model inductances can be derived directly by analyzing the network:

$$L_1 = L_X + M = 3 + 4 = 7\,\text{H}$$

$$L_2 = L_Y + M = 2 + 4 = 6\,\text{H}$$

$$\boxed{A)\, L_1 = 7\,\text{H}, \quad L_2 = 6\,\text{H}, \quad L_3 = -4\,\text{H}}$$

Q1.24.1 NAT 2024 2M Ans: 0.25 ●●○

Let the two self-inductances be equal such that:

$$L_1 = L_2 = L$$

When connected in a series-aiding configuration, the equivalent total inductance is given by:

$$L_1 + L_2 + 2M = 500\,\text{mH} \implies 2L + 2M = 500\,\text{mH} \quad \text{— (1)}$$

When connected in a series-opposing configuration, the equivalent total inductance is given by:

$$L_1 + L_2 - 2M = 300\,\text{mH} \implies 2L - 2M = 300\,\text{mH} \quad \text{— (2)}$$

Adding equations (1) and (2):

$$2(L_1 + L_2) = 800\,\text{mH} \implies 2(2L) = 800\,\text{mH}$$

$$L = L_1 = L_2 = 200\,\text{mH}$$

Substituting $L = 200\,\text{mH}$ back into equation (1):

$$2(200) + 2M = 500 \implies 2M = 100 \implies M = 50\,\text{mH}$$

The coupling coefficient K is calculated as:

$$K = \frac{M}{\sqrt{L_1 L_2}} = \frac{M}{L} = \frac{50\,\text{mH}}{200\,\text{mH}} = \frac{1}{4} = 0.25$$

$$\boxed{0.25}$$

Q1.23.1 NAT 2023 2M Ans: 17.37

Given,

$$K = 0.6$$

$$L_1 = 4\,\text{H}, \quad L_2 = 5\,\text{H}$$

$$i_1(t) = 1\sin(200\pi t)\,\text{mA} \implies I_{1P} = 1\,\text{mA}$$

$$i_2(t) = 2\sin(200\pi t)\,\text{mA} \implies I_{2P} = 2\,\text{mA}$$

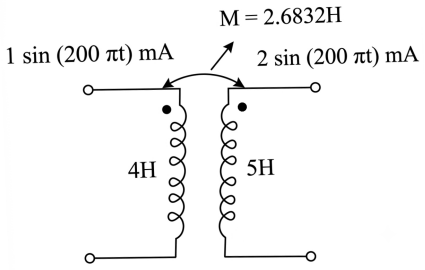


Fig. Solution Q1.23.1

Mutual inductance M:

$$M = K\sqrt{L_1 L_2} = 0.6\sqrt{4 \times 5} = 0.6\sqrt{20} \approx 2.6832\,\text{H}$$

The maximum peak energy stored in the mutually coupled circuit occurs when both currents reach their peak values simultaneously (since they are in phase). Looking at the dot polarity, both currents enter the dots, meaning the magnetic fields aid each other:

$$E_{\text{Peak}} = \frac{1}{2}L_1 I_{1P}^2 + \frac{1}{2}L_2 I_{2P}^2 + M I_{1P} I_{2P}$$

Substituting peak values in milliamperes ($1\,\text{mA} = 10^{-3}\,\text{A}$):

$$E_{\text{Peak}} = \frac{1}{2}(4)(1 \times 10^{-3})^2 + \frac{1}{2}(5)(2 \times 10^{-3})^2 + 2.6832(1 \times 10^{-3})(2 \times 10^{-3})$$

$$E_{\text{Peak}} = \left[2 \times 10^{-6}\right] + \left[10 \times 10^{-6}\right] + \left[5.3664 \times 10^{-6}\right]\,\text{J}$$

$$E_{\text{Peak}} = (2 + 10 + 5.3664)\,\mu\text{J} = 17.3664\,\mu\text{J}$$

Rounding off to two decimal places:

$$\boxed{17.37}$$

Key Points

- Total instantaneous energy stored in coupled coils: $E(t) = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2$
- The sign of the M term is positive if both currents enter (or both exit) the dot terminals.

Mistakes to be avoided

- Do not forget to convert milliamperes to amperes ($\times 10^{-3}$), which becomes 10^{-6} (micro) when squared or multiplied.

Q1.15.1 MCQ 2015 1M Ans: B ●○○

From the ideal transformer voltage ratio relations:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$\frac{V_2}{V_1} = \frac{1}{N}$$

$$V_2 = NV_1 \implies NV_i \sin \omega t$$

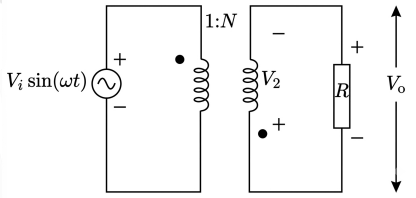


Fig. Solution Q1.15.1

According to the given dot polarity convention, the current enters the dot on the primary winding while the induced voltage is positive at the terminal opposite to the dot on the secondary winding. Thus, the polarity of V_2 is inverted relative to the marked terminal orientation of V_o:

$$V_o = -NV_i \sin \omega t$$

$$\boxed{B) -NV_i \sin \omega t}$$

Q1.13.1 MCQ 2013 2M Ans: B ●●●

We evaluate the circuit arrangement under two distinct excitation setups:

Case 1: $V_{wx1} = 100$ V is applied across WX terminals. The secondary side voltage of the ideal transformer with a turns ratio of 1 : 1.25 becomes:

$$V_{\text{secondary}} = 100 \times 1.25 = 125 \text{ V}$$

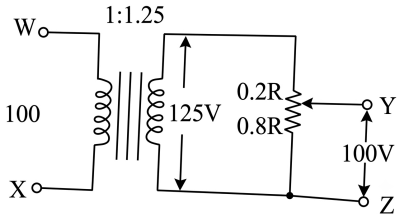


Fig. Solution Q1.13.1

The attenuator network splits the resistance into a $0.2R$ and $0.8R$ potentiometer configuration. The open-circuit voltage across the output terminals YZ is taken across the $0.8R$ section:

$$V_{yz1} = 125 \times \frac{0.8R}{0.2R + 0.8R} = 125 \times 0.8 = 100\,\text{V}$$

Hence, the first ratio is:

$$\frac{V_{yz1}}{V_{wx1}} = \frac{100}{100}$$

Case 2: $V_{yz2} = 100\,\text{V}$ is applied backward across YZ terminals. Since it is an ideal transformer, it draws zero magnetizing current to set up the rated flux ($i = 0$). Consequently, there is no current flow and no voltage drop across the upper attenuator resistor segment $0.2R$:

$$V_{secondary} = V_{yz2} = 100\,\text{V}$$

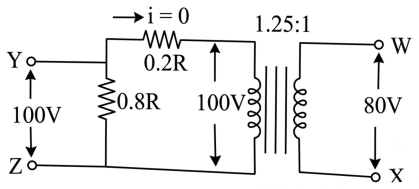


Fig. Solution Q1.13.1

Stepping down the voltage to the primary winding using the effective 1.25 : 1 reverse turns relation:

$$V_{wx2} = \frac{100}{1.25} = 80\,\text{V}$$

Hence, the second ratio is:

$$\frac{V_{wx2}}{V_{yz2}} = \frac{80}{100}$$

$$\boxed{B)100/100\ and\ 80/100}$$

Q1.13.1 MCQ 2013 2M Ans: C ● ○ ○

The individual quality factors (Q-factors) for the two uncoupled inductive coils at a chosen operating angular frequency ω for a series RL circuit are defined as:

$$q_1 = \frac{\omega L_1}{R_1}, \quad q_2 = \frac{\omega L_2}{R_2}$$

When these two coils are connected in series, the total equivalent inductance L_t and total equivalent resistance R_t become:

$$L_t = L_1 + L_2$$

$$R_t = R_1 + R_2$$

The effective quality factor q_t of the series combination at the same operating frequency is given by:

$$q_t = \frac{\omega(L_1 + L_2)}{R_1 + R_2}$$

Substituting $\omega L_1 = q_1 R_1$ and $\omega L_2 = q_2 R_2$ into the equation:

$$q_t = \frac{q_1 R_1 + q_2 R_2}{R_1 + R_2}$$

$$\boxed{C)(q_1 R_1 + q_2 R_2)/(R_1 + R_2)}$$

SOLUTIONS

II

Single Phase Transformers

Topics

1. Equivalent Circuit and Phasor Diagram
2. Open Circuit and Short Circuit Tests
3. Voltage Regulation and Efficiency of Transformer

Unit Snapshot*

Stream: IN	**Questions:** 7	**MCQ:** 2	**MSQ:** 0	**NAT:** 5
1M: 2	**2M:** 5	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2025)–3M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M Ans: C ● ○ ○

The secondary impedance referred to the primary side is:

$$Z_2' = \frac{Z_2}{K^2}, \quad \left(K = \frac{N_2}{N_1}\right)$$

$$Z_2' = \left(\frac{N_1}{N_2}\right)^2 \cdot Z_2$$

For a step-down transformer:

$$\frac{N_1}{N_2} > 1$$

$$\therefore Z_2' > Z_2$$

$$\boxed{C) Z_2' > Z_2}$$

Q2.26.2 NAT 2026 2M Ans: 2490 to 2510 ● ● ○

Given parameters:

$$\cos\phi = \frac{P}{VI} = \frac{363}{220 \times 2.75} = 0.6$$

The magnetizing current component is:

$$I_m = I_0 \sin\phi = 2.75 \times 0.8 = 2.2\,\text{A}$$

The magnetizing reactance X_m relative to the low-voltage side is:

$$X_m = \frac{V}{I_m} = \frac{220}{2.2} = 100\,\Omega$$

Referred to the high-voltage side, the magnetizing reactance becomes:

$$X_m' = \frac{X_m}{K^2} = \frac{100}{\left(\frac{220}{1100}\right)^2}$$

$$= \left(\frac{1100}{220}\right)^2 \times 100 = 2500\,\Omega$$

$$\boxed{2500}$$

Q2.25.1 NAT 2025 2M Ans: 31.00 to 31.50 ● ● ○

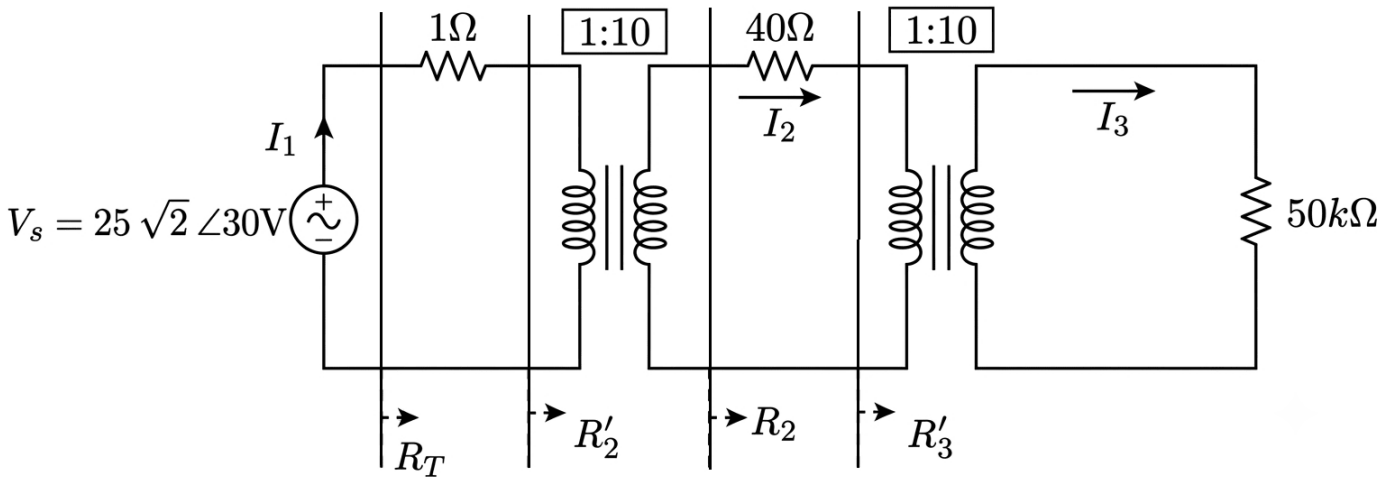


Fig. Solution Q2.25.1

The transformer network can be systematically referred across stages by squaring the respective turns ratios. Referring the loads back step-by-step simplifies the multi-stage network into a single equivalent loop:

$$R_{\text{eq}} = 1\,\Omega + 4\,\Omega = 5\,\Omega$$

The loop current is found using the RMS voltage of the source:

$$I = \frac{25}{10}\,(\text{rms}) \implies I = 2.5\,\text{A}$$

The power dissipated in the load is:

$$P_{50\,\text{k}\Omega} = P_{5\,\Omega} = 2.5^2 \times 5 = 31.25\,\text{W}$$

$$\boxed{31.25}$$

Q2.24.1 NAT 2024 2M Ans: 95.2 to 95.4 ●●○

Given parameters: Rating = 50 kVA, $\eta_{\text{FL, UPF}} = 95\%$, pf = 1 and Core losses = neglected ($W_i = 0$)
Using the full-load efficiency equation at unity power factor:
$W_{\text{cu}}(x) = x^2 W_{\text{cu, FL}}$

$$\text{Efficiency} = \frac{Output\ Power}{Output\ Power + Iron\ Loss + Copper\ Loss}$$

$$\%\eta_{\text{FL, UPF}} = \frac{50 \times 10^3 \times 1 \times 1}{(50 \times 10^3 \times 1 \times 1) + (W_i = 0) + W_{\text{cu, FL}}}$$

$$0.95 = \frac{50 \times 10^3}{(50 \times 10^3) + W_{\text{cu, FL}}} \implies W_{\text{cu, FL}} = 2631.57\,\text{W}$$

Now computing efficiency at 75% full load and 0.8 power factor lagging:

$$\%\eta_{75\%\text{FL},0.8\text{pf}} = \frac{(0.75 \times 50 \times 10^3) \times 0.8}{(0.75 \times 50 \times 10^3 \times 0.8) + 0 + (0.75)^2 \times 2631.57} \times 100$$

$$= 95.29\%$$

$$\boxed{95.2\ to\ 95.4}$$

Q2.23.1 NAT | 2023 | 2M | Ans: 1 | ● ● ●

Given short-circuit test data at $f = 1\,\text{kHz}$: $w_{sc} = 8\,\text{W}, V_{sc} = 6\,\text{V}, I_{sc} = 2\,\text{A}$

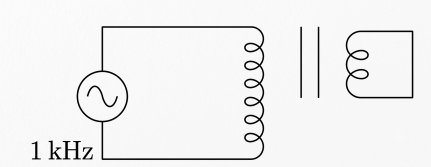


Fig. Solution Q2.23.1

The short-circuit impedance and equivalent parameters referred to the primary are:

$$Z_{sc} = \frac{V_{sc}}{I_{sc}} = \frac{6}{2} = 3\,\Omega$$

$$w_{sc} = I_{sc}^2 R_{01} \implies R_{01} = \frac{8}{2^2} = 2\,\Omega$$

$$Z_{sc}^2 = R_{01}^2 + X_{01}^2 \implies X_{01}^2 = 3^2 - 2^2 = 5$$

$$X_{01} = \sqrt{5} \implies X_{01} = +j\sqrt{5}$$

When connected with a capacitor C across a 2 V source:

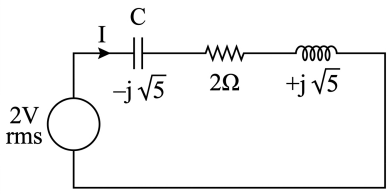


Fig. Solution Q2.23.1

The capacitance value is:

$$C = \frac{1}{2\pi\sqrt{5}}\,\text{mF}$$

The capacitive reactance is:

$$-jX_c = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 10^3 \times \frac{1}{2\pi\sqrt{5}} \times 10^{-3}} = -jX_c = \sqrt{5}$$

Evaluating the final circuit loop at resonance: Since $X_L = X_C$, the circuit is purely resistive and under a resonance condition:

$$I = \frac{V}{R} = \frac{2}{2} = 1\,\text{A}$$

$$\boxed{1}$$

Q2.21.1 NAT 2021 1M Ans: 0.24 to 0.26 ● ● ○

Given parameters: $L_0 = 250 \times 10^{-3}$ H , $R_0 = 300\,\Omega$ The inductive reactance is:

$$X_{L_0} = 2\pi f L_0 = 2\pi \times 50 \times 250 \times 10^{-3} = 78.54\,\Omega$$

The susceptance and conductance are:

$$B_{L_0} = \frac{1}{X_{L_0}} = \frac{1}{78.54} = 0.01273\,\text{℧}$$

$$G_0 = \frac{1}{300} = 3.33 \times 10^{-3}\,\text{℧}$$

The total admittance in parallel circuit is:

$$Y_0 = \sqrt{G_0^2 + B_{L_0}^2} = 0.01316\,\text{℧}$$

The no-load power factor is:

$$\cos\phi_0 = \frac{G_0}{Y_0} = 0.253 \quad \text{(lag)}$$

$$\boxed{0.253}$$

Q2.21.2 MCQ 2021 2M Ans: A ● ○ ○

Given parameters: $\eta_{\max} = 98\%$, $W_{\text{iron}} = 80\,\text{W}$, $R_{01} = 0.5\,\Omega$, $I_1 = 25\,\text{A}$ At maximum efficiency, the variable copper loss equals the constant iron loss (x is the loading fraction):

$$x^2 I_1^2 R_{01} = W_i$$

$$x^2 (25)^2 (0.5) = 80$$

$$x = 50.59\%$$

Thus, maximum efficiency occurs at 50.59% of full load.

$$\boxed{A)50.6\%}$$

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SOLUTIONS

III

Three Phase Induction Machines

Topics

1. Principle of Operation , Types and Performance
2. Equivalent Circuit of Induction Motor
3. Torque-Slip, Torque-Speed Characteristics
4. Stability and Testing – No-Load (OC) and Blocked-Rotor (SC) Tests
5. Starting of Induction Motors
6. Speed Control
7. Types of losses and efficiency calculation of Electric Machines

Unit Snapshot*

Stream: IN	**Questions:** 5	**MCQ:** 2	**MSQ:** 0	**NAT:** 3
1M: 2	**2M:** 3	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2023)–2M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q3.25.1 MCQ 2025 2M Ans: A ●●○

Given conditions:

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{2} = 3000\,\text{rpm}$$

The initial slip is:

$$s_1 = \frac{3000 - 2970}{3000} = 0.01$$

In the normal operating (low-slip) region of a standard induction motor, torque varies linearly with slip.

$$T_1 \propto s_1$$

As torque is doubled, $T_2 = 2T_1$, meaning the slip gets doubled as well:

$$S_2 = 2S_1 = 2 \times 0.01 = 0.02$$

Evaluating the new rotor speed N_2:

$$S_2 = \frac{N_s - N_2}{N_s} \implies \frac{3000 - N_2}{3000} = 0.02$$

$$3000 - N_2 = 60 \implies N_2 = 2940\,\text{rpm}$$

A) 2940

Q3.24.1 NAT 2024 2M Ans: 17.00 to 17.30 ●●○

Given:

$$T_{st} = 100\% \text{ of } T_{FL} \implies \frac{T_{st}}{T_{FL}} = 1$$

$$T_{max} = 300\% \text{ of } T_{FL} \implies \frac{T_{max}}{T_{FL}} = 3$$

$$S_{T\,max} = ?$$

The ratio of starting torque to maximum torque is a specific function of the parameter a, where a corresponds to the slip at which maximum torque occurs (s_m).

Using the torque ratios:

$$\therefore \frac{T_{st}/T_{FL}}{T_{max}/T_{FL}} = \frac{1}{3} \implies \frac{T_{st}}{T_{max}} = \frac{1}{3} = \frac{2a}{1 + a^2}$$

$$\frac{2a}{1 + a^2} = \frac{1}{3} \implies a^2 + 1 - 6a = 0$$

$$a^2 - 6a + 1 = 0$$

Solving for a (slip at maximum torque):

$$a = 0.1715 = 17.15\%$$

17.15

Q3.23.1 NAT 2023 1M Ans: 0 ●●○

Output voltage when machine is rotated in the clockwise direction is 200 V. If it is rotated in the opposite direction, the armature current direction and shunt field current direction will be changed. Then ϕ_{sh} opposes the $\phi_{residual}$. Net EMF will become zero.

0

Q3.22.1 NAT 2022 1M Ans: 1440 ● ○ ○

Given parameters: $V = 440\,\text{V}$, $P = 8\,\text{kW}$, Poles $= 4$, $f = 50\,\text{Hz}$, $S = 0.04$ The synchronous speed is calculated as:

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

The actual rotor speed N_r is:

$$N_r = N_s(1 - s) = 1500(1 - 0.04) = 1440\,\text{rpm}$$

1440

Q3.21.1 MCQ 2021 2M Ans: A ● ● ○

By adding external resistance in the rotor circuit of a slip ring induction motor, the total rotor resistance increases, which subsequently shifts the maximum torque point and increases the starting torque. This improve the starting performance of a wound-rotor induction motor by bringing the starting torque closer to the maximum available breakdown torque.

A) The starting torque would be higher.

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